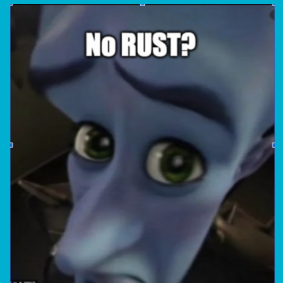


OXIDATION-REDUCTION REACTIONS-Luigi Brugnatelli and The Development of Electroplating(62-66)

By Matija Benko

What is Oxidation?

- Oxidation frees electrons from one reactant, then are used in the reduction of other reactant
 - sodium has 11 electrons, one is valence electron; when sodium changes to an ion with +1 charge, it frees an electron



What is Reduction?

- The freed electron from Na goes into a positive Ag ion, making the latter turn into a Ag atom(reduction)
- Oxidation-reduction(Redox) reaction is one or more electrons getting sent between reactants

knowing reduction means
negative ions gaining electrons



>:)

Me before I
knew about redox



;-;

Oxidation Numbers

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- Oxidation # is # of electrons an atom has to get/give to separate from a compound
 - Can be positive/negative/zero
 - In AgCl, Ag is +1 and Cl is -1
 - in Al_2O_3 , each Al atom # is +3, each O atom # is -2

Rules for Oxidation Numbers

- There are two regular* rules for knowing compounds that form these elements:
 - H and group 1 elements have oxidation #s of +1
 - The oxidation # of O is -2

FIGURE 68

1 1A																	18 8A
1 H +1 -1	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	10 Ne
3 Li +1	4 Be +2											5 B +3	6 C +4 +2 -4	7 N +5 +4 +3 +2 +1 -3	8 O +2 -2 -1 -2	9 F -1	
11 Na +1	12 Mg +2											13 Al +3	14 Si +4 -4	15 P +5 +3 -3	16 S +6 +4 +2 -2	17 Cl +7 +6 +5 +4 +3 +2 +1 -1	18 Ar
		3 3B	4 4B	5 5B	6 6B	7 7B	8	9	10	11 1B	12 2B						
19 K +1	20 Ca +2	21 Sc +3	22 Ti +4 +3 +2	23 V +5 +4 +3 +2	24 Cr +6 +5 +4 +3 +2	25 Mn +7 +6 +5 +4 +3 +2	26 Fe +8 +7 +6 +5 +4 +3 +2	27 Co +9 +8 +7 +6 +5 +4 +3 +2	28 Ni +10 +9 +8 +7 +6 +5 +4 +3 +2	29 Cu +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	30 Zn +12	31 Ga +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	32 Ge +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	33 As +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	34 Se +16 +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	35 Br +17 +16 +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	36 Kr
37 Rb +1	38 Sr +2	39 Y +3	40 Zr +4	41 Nb +5 +4	42 Mo +6 +5 +4 +3	43 Tc +7 +6 +5 +4	44 Ru +8 +7 +6 +5 +4 +3	45 Rh +9 +8 +7 +6 +5 +4 +3 +2	46 Pd +10 +9 +8 +7 +6 +5 +4 +3 +2	47 Ag +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	48 Cd +12	49 In +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	50 Sn +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	51 Sb +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	52 Te +16 +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	53 I +17 +16 +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	54 Xe
55 Cs +1	56 Ba +2	57 La +3	72 Hf +4	73 Ta +5	74 W +6 +5 +4	75 Re +7 +6 +5 +4	76 Os +8 +7 +6 +5 +4	77 Ir +9 +8 +7 +6 +5 +4 +3	78 Pt +10 +9 +8 +7 +6 +5 +4 +3 +2	79 Au +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	80 Hg +12	81 Tl +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	82 Pb +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	83 Bi +15 +14 +13 +12 +11 +10 +9 +8 +7 +6 +5 +4 +3 +2 +1	84 Po +16	85 At -1	86 Rn

Possible oxidation numbers for the elements.

Image from Chemistry, 7th edition, by Raymond Chang, published by McGraw-Hill, © 2002.

Rules for Oxidation Numbers

More rules for oxidation (part #2) in general:

- Oxidation # of an ion is its charge
- Neutral atoms have an oxidation # of 0
- Oxidation #s of atoms in neutral compounds must add up to 0 (neutral compounds have no charge)
- Sum of the oxidation #s of atoms inside ions made with atoms must be equal to the charge on said ion
- These rules can turn a chemical equation into an Algebraic one



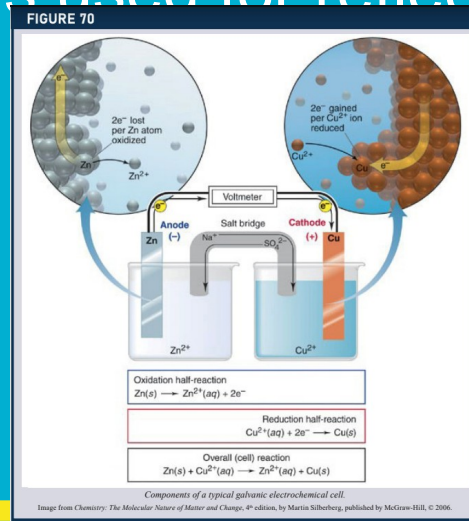
The Role of the Electron in Oxidation-Reduction

- # of electrons freed during oxidation must equal # of electrons added during reduction
 - Redox reaction equation use Half reaction, which shows the Oxidation and Reduction in two equations
 - Half reaction of $\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$
 - 1. $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
 - 2. $2\text{Ag}^+ + 2\text{e}^- \rightarrow 2\text{Ag}$



ELECTROCHEMISTRY

- Electrochemistry the study of chemical processes that cause electricity
 - Chrome plating is used for reflective surfaces on cars and appliances



Cell potentials



- Volts: How much an electron wants to get away/join an atom/ion
 - Cell potential: potential energy for a redox reaction(measured in Volts)
 - Batteries has cell potential
 - When all of the reaction in a cell potential has been done, there is no more potential energy left
 - Each half reaction can be assigned a potential
 - Standard half-cell potentials are measured relative to $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

Explanation of Cell potentials

- Oxidation is just the reverse of reduction, so sign of the volts are opposites
 - if the oxidation reaction is $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$, then the standard oxidation potential of the half-cell is – 0.34V(Volts)
- Half-cell potentials are combined to provide information on a complete reaction
 - $E^\circ_{\text{red}} - E^\circ_{\text{ox}}$

Example of Cell

potentials

For the reaction, $\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$, voltage of the reaction is determined by first identifying the potential of each half reaction:

- $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ $E^0 = -0.34 \text{ V}$
- $\text{Ag}^+ + 1\text{e}^- \rightarrow \text{Ag}$ $E^0 = +0.80 \text{ V}$
 - Sign of the Cu half-reaction voltage was changed since it was the oxidizing half-reaction
- Add the voltage of both half reactions to get the voltage of the redox reaction
 - $-0.34\text{V} + 0.80\text{V} = 0.46\text{V}$ (spontaneous occurring)

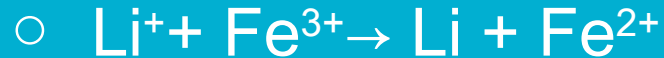


Example of Cell

potentials

Spontaneously occurring: found often in nature

- Non-spontaneously occurring: not found often in nature



- The two half-reactions are as follows



- Of course, the final result is a negative number (because $-n-m$ is negative)

- This reaction happens naturally in reverse, so the result would be positive



**Thank
You!**